Model for the Implementation of Digital Services in the Business Processes of Higher Education Institutions Under Martial Law

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Abstract. The paper examines approaches to organizing the educational process in Ukrainian higher education institutions during martial law, emphasizing the integration of modern information and communication technologies with a pedagogical focus. Virtual learning environments and learning management systems have become essential for maintaining educational continuity despite unprecedented challenges. The impact of war on students' general conditions and learning experiences has been analyzed, revealing that these digital platforms offer flexibility and accessibility for both students and faculty. They enable academic activities to continue despite physical displacement and disruptions. However, the rapid transition to online learning has exposed complexities in integrating educational software, particularly the need for students and academic staff to develop the necessary digital skills. Digital literacy gaps and technical support challenges have been identified as significant barriers to the full utilization of these platforms. Addressing these issues requires targeted measures to enhance digital competencies, ensuring participants can effectively navigate virtual learning environments.

Based on the experience of Kherson State University, several initiatives have been proposed to improve digital literacy and ensure the effective use of educational platforms. Establishing dedicated digitalization assistant positions, implementing comprehensive training programs, and providing continuous technical support are key measures to facilitate this transition. The integration of artificial intelligence assistants can further enhance accessibility by offering guidance when human support is unavailable. Findings suggest that a combination of technological infrastructure and pedagogical support is crucial for the long-term resilience of higher education in Ukraine. Continued investment in digital skills development will help mitigate future disruptions, ensuring that higher education institutions can maintain stability and adaptability in crisis conditions.

Keywords: distance learning · educational process · virtual learning environments · learning management systems

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1 Introduction

1.1 Problem Statement

The Russian aggression that began in 2022 has posed unprecedented challenges for Ukraine's higher education system. Higher, secondary, school, and preschool education institutions in the most affected regions—Luhansk, Donetsk, Kharkiv, Zaporizhzhia, and Kherson—have faced severe disruptions. The occupation of territories, destruction of infrastructure, and security threats to students, academic staff and scientists have necessitated urgent solutions to sustain educational and business processes within higher education institutions. One of the key solutions to the crisis was the relocation of universities to relatively safer regions, requiring not only the physical transfer of material assets but also significant transformations in learning formats, educational process management, and support for students and academic staff.

Information and communication technologies play a fundamental role in organization and maintaining the education processes during wartime. Modern software and technological systems give higher education institutions the ability to establish effective communication among participants, manage and regulate access to educational materials, and implement key elements of remote and blended learning.

The virtual learning environments (VLE) and learning management systems (LMS) based on cloud technologies had became a crucial tools aimed for:

- mitigating the consequences of forced displacement through international collaboration;
- enhancing the resilience of the educational process to unpredictable disruptions.

Partnership projects with leading global technology companies such as Microsoft, Google, and Amazon Web Services (AWS) have facilitated the effective integration of modern digital platforms and tools. At Kherson State University, this collaboration has accelerated the development of its proprietary virtual learning environment, KSU24, leveraging the university's extensive experience in development of pedagogical software and integrated learning environments.

The implementation of virtual learning environments, learning management systems, and the organization of training processes for academic staff, administrative personnel, and students has helped to mitigate the impact of forced displacement. However, the complexity of this integration process has necessitated the introduction of a new university position—the **"digitalization assistant"**, whose responsibilities focus on assisting academic staff and students in mastering new ICT tools and components of the virtual learning environment.

Thus, the preservation of higher education in Ukraine during wartime requires not only the physical relocation of universities and digital transformation but also maximum support for academic staff and students in mastering new information and communication technologies. This approach enables higher education institutions to successfully function in the new realities and maintain their key mission—providing quality education.

1.2 Review of Recent Research and Publications

Virtual learning environments have become one of the mainstream approaches in education all over the world over the last decades. As stated in [1]: "a Virtual Learning Environment (VLE) is an accumulation of incorporated software components, empowering the administration of e-learning. The quality of VLE is indispensable for the accomplishment of e-learning goals and responsible to engage students during learningâĂİ. The combination of software components of virtual learning environments allow the support of all forms of education that are mostly suitable for the concrete educational requirements. Some universities in Ukraine were forced to transfer their educational process to online formats [2,3], using the corporate software instruments [4]. Following the fast transition the high education institutions and their fellow software companies had to start the crisis overcoming strategies [5,6]. Especially hard this transition hit the relocated universities [7], that were forced additionally to move their entire infrastructure to safer regions [8]. All these led to not only digital transformation of educational processes but also to additional difficulties for pedagogical staff, who were themselves forced to adapt to new conditions as quickly as possible [9], and to adapt the existing educational programs and learning materials [10]. Also educational institutions had to face unexpected troubles, related to regional features that affect both students and teachers [11]. All these transformations and efforts were taken in order to minimize the negative impact of war [12] to provide the sustainable functioning of educational institutions [13]. AI assistants integrated into the virtual learning environment also provide a decent level of support for both educators and students, crucial to integrate insights from the study [14] to ensure the effective implementation of AI-driven policies for learning, teaching, and research.

1.3 The Research Goal

The goal of this research is to explore solutions for ensuring the continuity of higher education during wartime while enhancing the safety and quality of educational services through the adoption of educational staff and students to new technologies.

To achieve this goal, the following objectives must be addressed:

1. The most suitable approach for the education process organisation has to be chosen, based on the existing experience of relocated universities. In light of recent disruptions, particularly due to wartime conditions, universities must adapt swiftly to ensure that the education process continues with minimal interruption.
2. The most suitable software systems have to be integrated into the educational and business processes of concrete universities. The integration of virtual learning platforms and software systems plays a crucial role in facilitating the continuity of education under crisis conditions. Universities must identify and implement software systems that not only support online learning but

also streamline administrative, communication, and support services. These systems should include robust Learning Management Systems (LMS), student information systems, and communication tools that facilitate real-time interaction between faculty and students.

3. It is necessary to develop new approaches and methods for assessing the effectiveness of the educational process that take into account the specificities of organizing the educational process in a displaced higher education institution, as well as the challenges posed by martial law. Existing assessment models, which often rely on face-to-face interactions, in-person evaluations, and established physical infrastructures, may not be fully applicable in contexts where universities have been relocated or have adopted predominantly virtual learning environments. As such, new evaluation frameworks must be developed that reflect the unique challenges and opportunities of displaced institutions.

2 The Digital Services Implementation in Higher Education Institutions During Martial Law

2.1 Business Process Criterias in Virtual Learning Environments of Relocated Universities

The KSU24 Virtual Learning Environment is seamlessly integrated with all corporate accounts available to students. To ensure an optimal user experience for common students or teachers of the relocated university that has to face similar challenges as Kherson state university, the following essential criteria has to be established for the university business processes model (Fig. 1).

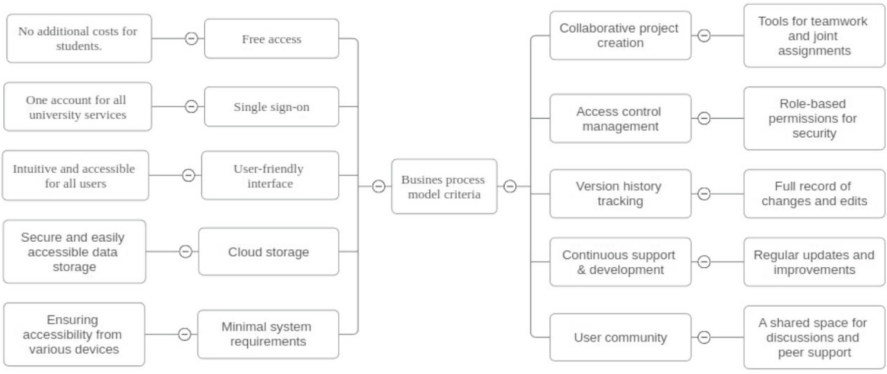


Fig. 1. Mind map for digital services for university during martial law key requirements model.

Free Access Criteria. Providing free access to the educational environment is crucial for ensuring equal learning opportunities for all students, especially in the context of a relocated university in Ukraine.

Single Sign-On. A unified account system simplifies access to various educational tools and platforms within the digital infrastructure of relocated universities.

User-Friendly Interface. A well-designed, easy-to-use interface is essential for students and educators who may not be technologically proficient.

Cloud Storage. Cloud-based data storage allows students and faculty staff to access course materials, assignments, and research files from any location, which is critical for a relocated university where physical access to campus resources may be limited.

Minimal System Requirements. Students from a relocated university may be using different types of devices, including older laptops, tablets, or even smartphones.

Access Control Management. In an educational environment, different users require different levels of access. Professors, teachers, students, and administrative staff must have clearly defined permissions to ensure data security and prevent unauthorized modifications to important materials.

Version History Tracking. With students and instructors frequently working on shared documents, version history tracking ensures that no progress is lost due to accidental deletions or overwrites.

Continuous Support. As the needs of a relocated university evolve, the KSU24 platform must continue to adapt and improve.

User Community. A strong online community helps students stay connected, share knowledge, and support each other through academic and personal challenges.

2.2 Learning Process Quality Assurance Instruments and Models

Another main issue that education institutions have to face during martial law - is the quality assurance of all the educational and administrative processes. This process becomes the crucial part of the relocated university educational and business processes. The KSU24 Virtual Learning Environment integrates technologies that currently allows the following automated reports:

- *Student and Staff Location Report* – Displays the geographical distribution of students and staff, which is crucial for a relocated university.
- *Student Attendance Percentage Report* – Provides detailed insights into student participation in classes.
- *Final Ranking Report of Academic Staff* – Summarizes the performance and achievements of academic staff.
- *Synchronous vs. Asynchronous Class Sessions Report* – Analyzes the distribution of live and recorded sessions.
- *Survey Results from Closed Course Logs* – Gathers and presents feedback from students based on completed courses.
- *Statistics on Safety Protocol Acknowledgment* – Tracks the completion of safety training and compliance with university regulations, ensuring that students and staff are aware of required protocols.
- *Statistics on Signing the Academic Integrity Code* – Monitors student and faculty commitments to academic integrity.

The system also allows the manual creation of any complicated report based on the data present in the virtual learning environment database. The present report creation and generation system is used for processes of advanced data selection and business process optimization, powered by Pandas library, provide the capability to generate customized data selections based on any specified criteria. From the system administrator point of view the creation of a new report is no different than writing a quarry in a SQL-line database. The example view of the page containing the specialized editor for creation of the new simple report is shown on Fig. 2.

The report presented on Fig. 2 outputs the human resources (student, teachers, administrative staff) registration info. On the upper part of the window the user that has required access level prints the data he wants to be present in his report. In the current query the user entered that he wants three fields from the database: the unique id of a person, the address of person registration, and the creation date.

By providing real-time, data-driven insights, the reporting system in KSU24 plays a critical role in enhancing transparency, optimizing workflows, and improving overall university management. The ability to dynamically generate reports based on specific needs ensures that educational processes remain efficient, responsive, and aligned with the evolving requirements of a modern, relocated university.

2.3 Role of the Digitalization Assistant Position in the Educational Processes in Relocated University

In the modern educational landscape, digital transformation plays a crucial role in ensuring the continuity of academic processes, especially for universities that have been forced to relocate. The *digitalization assistant* is a specially designated position within Kherson State University that serves as a bridge between faculty, students, virtual learning environment, and the other software services

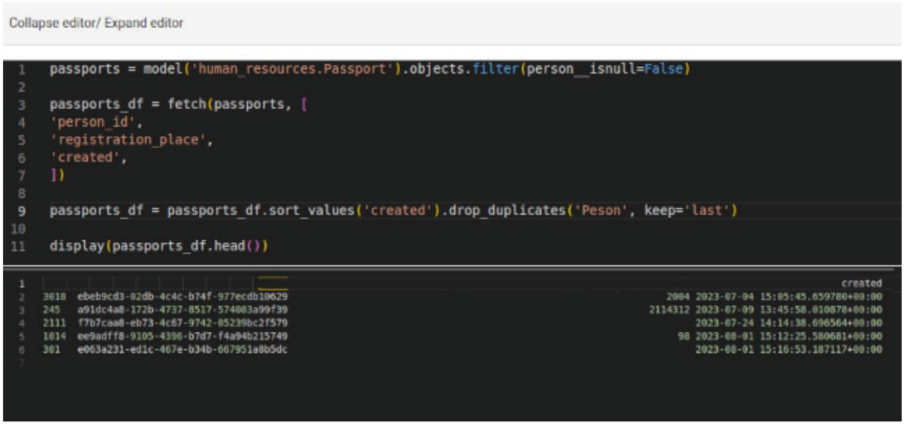


Fig. 2. The editor window for creating and executing custom reports.

available in university. This role is fundamental in simplifying the transition to online education, especially during the war crisis that significantly forced this transition. *Digitalization assistant* provides technical and methodological support, and ensures that all members of the university community can efficiently utilize new digital tools.

Beyond technical support, the digitalization assistant contributes to the optimization of administrative processes within a relocated university. With the integration of digital reporting systems powered by Pandas, universities can now track student attendance, generate faculty performance reports, manage online course statistics, and oversee compliance with institutional policies.

2.4 Role of the Student Feedback System in the Business and Educational Processes in Relocated Universities

In the context of a relocated university, a well-structured student feedback system plays a pivotal role in shaping educational strategies, improving teaching quality, and addressing student concerns. At Kherson State University, the current feedback system is developed as a module for the virtual learning environment KSU24, and is designed to collect responses from students after every lesson, at the end of each semester, and after the completion of the academic year. The general tab view on the statistics on the quizzes, presented on Fig. 3, shows the statistical data for the following questions: number of classes conducted by teachers, number of surveys completed by students, how convenient was feedback and communication with the teacher for students, how clear was the educational material for students, how effective are the approaches to activating cognitive interest and thinking activity for students, how interested students were in the topic and content of the class, how satisfied students are with the atmosphere in the class.

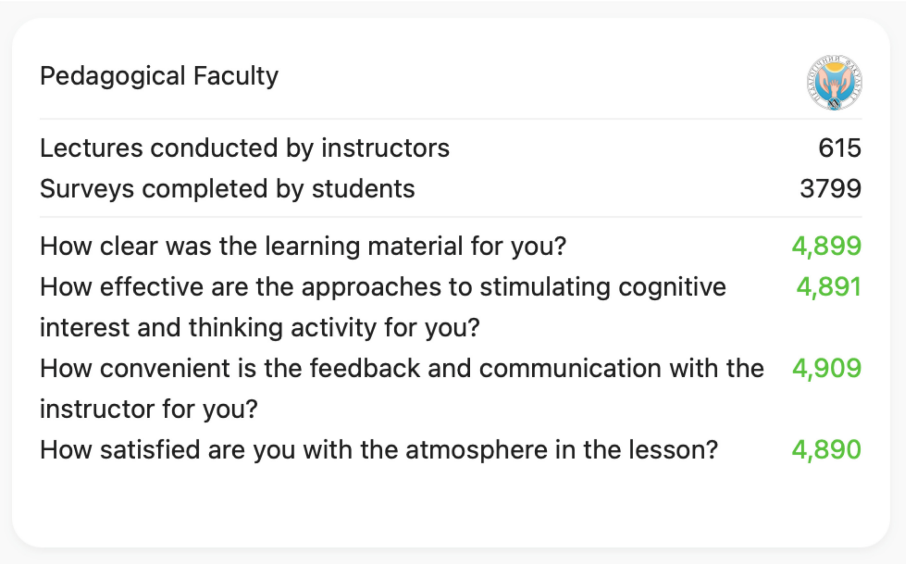


Fig. 3. The tab showing the general quizzes information about Pedagogical faculty.

In the context of a relocated university, where many students may experience challenges related to displacement, adjusting to online or hybrid learning formats, and coping with external stressors, the feedback system serves as a critical monitoring tool. Lastly, from an institutional perspective, the student feedback system is crucial for continuous improvement and decision-making. The data collected allows the university to make informed decisions on curriculum adjustments, teaching strategies, and administrative policies.

2.5 Role and Perspective of the Integration of AI-Based Assistants in the Business and Educational Processes in Relocated Universities

The integration of an AI assistant into the KSU24 Virtual Learning Environment presents significant pedagogical advantages, particularly in enhancing the effectiveness and accessibility of education for students and faculty in a relocated university. By analyzing data on student performance, engagement, and learning preferences, the AI assistant can recommend customized resources, suggest improvement strategies, and even provide instant feedback on assignments. This personalized learning approach can help bridge gaps in knowledge and ensure that students remain engaged in their learning journey, even in the face of external challenges like displacement or disruptions caused by the war. The general view of the AI assistant page is presented on Fig. 4., it is based on EVA-ai assistant [16].



Fig. 4. The KSU24 Ai assistant answering the required question.

One of the key roles of an AI assistant is its ability to facilitate active learning through adaptive learning algorithms. These algorithms enable the assistant to evaluate student inputs, identify areas where they struggle, and suggest supplemental materials or activities designed to reinforce their understanding. In addition, the AI assistant can support faculty members by streamlining administrative tasks and improving time management. From an administrative and management perspective, the AI assistant can play a crucial role in ensuring that students and faculty remain connected to the KSU24 Virtual Learning Environment. By facilitating communication between the university and its participants, the AI assistant can provide real-time updates on academic schedules, deadlines, and institutional news, ensuring that no student or faculty member misses important information. In summary, the integration of an AI assistant into the KSU24 Virtual Learning Environment provides profound pedagogical benefits, from offering personalized learning paths and supporting faculty to enhancing administrative efficiency and promoting digital literacy. Its ability to adapt to the needs of both students and faculty makes it a critical enabler of success in the transitional phase of the relocated university, ensuring that education continues to be effective, accessible, and resilient in these challenging times.

3 Conclusions

This paper has thoroughly explored the organizational approaches employed by higher education institutions in Ukraine during martial law based on the experience of Kherson state university. A significant outcome of the study is the realization that, while the transition to online learning platforms has facilitated continuity, it has also introduced challenges related to digital literacy and the technical competencies of both students and faculty. The paper highlights that the integration of educational software into existing institutional processes requires substantial efforts in training and capacity building. Without adequate

support, issues such as low digital literacy, technical barriers, and insufficient training can hinder the effective use of these platforms. The implementation of dedicated digitalization assistant positions and the integration of AI-powered assistants have been identified as essential strategies to bridge these gaps.

The research also emphasizes the importance of feedback systems in enhancing both pedagogical quality and administrative decision-making. The student feedback system implemented at KSU allows educators to continuously monitor student progress, identify academic challenges, and address potential learning obstacles in real-time. Looking forward, the paper suggests several avenues for further research to enhance the resilience and effectiveness of higher education in Ukraine, particularly in the context of relocated universities. One key area for future investigation is the scalability of virtual learning platforms and their potential application across a broader range of educational institutions in Ukraine. The integration of AI assistants, the development of comprehensive training programs, and the creation of digital literacy initiatives should be key components of these efforts.

In conclusion, the successful transition to virtual learning environments in Ukrainian higher education institutions during martial law highlights the importance of technology-driven solutions and pedagogical support. While challenges remain, particularly in the areas of digital literacy and technical support, the efforts to enhance these areas—through digitalization assistants, comprehensive training programs, and feedback systems—have proven to be essential in maintaining educational continuity and ensuring that students receive a high-quality education. The resilience demonstrated by Kherson State University and other relocated institutions is a testament to the power of digital tools, community support, and adaptability in overcoming the barriers posed by war. Moving forward, expanding these experiences to other educational facilities across Ukraine will be crucial for strengthening the long-term sustainability and effectiveness of higher education in the country during and beyond times of crisis.

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